Business Memo

A practical way to reduce COD returns without losing good customers

To: Founder **From:** Amaan Siddiqui, Associate Software Engineer Candidate **Date:** 30 October 2026
Subject: A balanced, order-by-order approach to COD risk

.

What you found in the data that surprised you (at least 3 real findings, with numbers)

I reviewed the supplied file covering 1 April to 30 September 2026. It has 2,018 rows. One order is still marked IN_TRANSIT, so I left it out of the model evaluation. That leaves 2,017 orders with a clear outcome: 1,444 delivered and 573 returned. The overall RTO rate is 28.4%.

- **First-time customers are riskier, but most are still genuine.** There were 460 RTOs among 1,517 first-time orders (30.3%). For repeat customers, it was 113 out of 500 (22.6%). So nearly 70% of first orders were delivered. Automatically cancelling first orders would mean turning away many customers who could become repeat buyers.
- **Meta Ads first orders deserve a check, not a blanket rejection.** This group had 196 RTOs from 572 orders (34.3%). That is higher than the overall rate, but about two out of every three still reached the customer. Since acquiring a new customer costs ₹300 or more, I would rather confirm the order or try a prepaid offer before cancelling it.
- **A repeated RTO history is a much stronger warning.** Customers with at least two previous RTOs had 26 returns from 34 orders (76.5%). I would treat this as a reason for closer review. The group is small, though, so I would not use it to justify blocking a whole location.

Your recommended rules, and what they would have done to last month's orders

I trained the JavaScript model on April-August orders and kept September separate as a test month. That gave 1,683 training orders and 334 September orders for the backtest. The model recorded a ROC-AUC of 0.739, a Brier score of 0.167 and average precision of 0.510. These results are encouraging for a prototype, but I would want more live data before relying on it for automatic decisions.

How you balanced Operations vs. Marketing, with the ₹ trade-off

I would start with a ₹20 discount and test it, rather than offer a large discount straight away. Using the assignment figures, the COD RTO rate is 28.4% and the prepaid RTO rate is 4%. Moving one order from the COD baseline to prepaid reduces expected RTO cost by about ₹41.33 (24.4 percentage points multiplied by ₹169.40). After the ₹20 discount, the estimated benefit is about ₹21.33 per successful conversion. If 35% of customers take the offer, that works out to roughly ₹7.47 per nudge on average, before payment fees and any change in repeat behaviour.

What could go wrong with your system and how we'd notice

- **Model quality can change.** I would review RTO rates and model performance every month, including separate views by courier, city, acquisition source and risk band. Small groups should always show their order count.

- **Our WhatsApp assumptions may be wrong.** The dataset does not tell us who replies, who pays prepaid or whether confirmation really reduces RTO. I would log every message and outcome, then replace the 70%, 50% and 35% assumptions with actual results.
- **A wrong address review can frustrate a good customer.** The optional AI check should only help identify whether an address looks complete. It should explain its concern and send uncertain cases to a person; it should never judge whether a customer is trustworthy.
- **A hold can turn into a lost sale if nobody follows up.** I would track how long orders stay on hold, how often an operator releases them and how many later deliver. A simple review-time target and a customer-service rescue step would help.

What you'd build next if you had 2 more weeks

Over the next two weeks, I would connect Shopify webhooks so each new order is scored automatically, then add an operator screen to release or reject a held order with a recorded reason. I would connect approved WhatsApp templates and a real payment confirmation webhook only after opt-in and privacy checks are in place. I would also start a small ₹10/₹20/₹30 discount test and measure conversion, RTO and repeat purchases before changing the offer.